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How can we enhance reputation in Blockchain consensus for Industry 4.0–A proposed approach by extending the PageRank algorithm

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ABSTRACT

This research presents a cost-effective approach toward an enhancement of Blockchain consensus targeting enhanced data security and privacy in the Industry 4.0 scenario by incorporating an enhanced reputation score at a cost-effective approach. Cellular Automata (CAs)-based PageRank validation (i.e., an extended PageRank) algorithm is incorporated towards consensus design in Blockchain Trilemma to facilitate cost-effectiveness in Industry 4.0 scenario. A detailed investigation is carried out to explore the true potential for CAs-based PageRank validation towards computation of Reputation score in Blockchain Trilemma. Further investigations and experimental analysis for the proposed approach towards computation of Reputation scores explore the efficiency of the proposed CAs-based PageRank validation-based approach over several existing approaches e.g., PageRank, NCDawareRank, and HodgeRank-based approaches. Additionally, energy efficiency in the proposed approach is ensured as low-energy consumption may be achieved with CAs-based physical modeling.

1. Introduction

The advances in the smart production lines in recent days may undoubtedly be concluded as a result that is achieved from systematic advancement and deployment of CPS (Cyber-Physical Systems). A synergetic integration of several manufacturing line components along with recent advancements in IT (Information Technology) and related technology tools have resulted in an effective and resilient performance in a properly distributed (collaborative) way in view of modern smart production lines which is widely acknowledged as the 4th industrial revolution (i.e., Industry 4.0). Several cutting-edge technologies e.g., Smart Sensors, Cloud Computing, IoT (Internet of Things), Cyber Security, Blockchain technologies, Big Data, Horizontal and Vertical System Integration, Simulation, Augmented Reality, Autonomous Robots, Autonomous Material Handling, Additive Manufacturing, etc. are united to accomplish the "the threshold of the fourth Industrial Revolution" (Lee et al., 2015). Thus, we found that the CPS architecture and Industry 4.0 are closely related (Lee et al., 2015; Mitra, 2021; Ahsan and Siddique, 2022; Lee et al., 2018; Jiang, 2018; Lee et al., 2014) and both are ever-evolving to achieve technology and quality advancements. Since CPS architecture in Industry 4.0 is at its early stage, several guidelines have been developed for its effective deployment in smart manufacturing lines (Lee et al., 2015; Mitra, 2021; Ahsan and Siddique, 2022; Lee et al., 2018; Jiang, 2018; Lee et al., 2014). Though several state-of-the-art pieces of literature focusing on several advancements in Industry 4.0 might be found in different scientific databases, we found that it was much impractical to report all those words in a single article. Hence, we chose to restrict our literature.

We found that the uses of CPS and Industrial AI (Artificial Intelligence) in Industry 4.0 were presented in (Lee et al., 2015; Lee et al., 2018; Jiang, 2018) to enhance the manufacturing line(s). Several other important concerns related to data processing, security, privacy, etc., in view of Industry 4.0 were addressed by researchers; for example, different Big Data processing were presented in Mitra (2021), Ahsan and Siddique (2022), Lee et al. (2014), and Blockchain technology-based security features were presented in (Viriyasitavat and Hoonsopon, 2019; Lu, 2019; Viriyasitavat et al., 2019; Zheng et al., 2017; Mohamed and Al-Jaroodi, 2019). Among the several areas in Industry 4.0, we were particularly interested in Blockchain-based security features in view of Industry 4.0 as it may play a significant role in the deployment of CPS in Industry 4.0 scenarios (Lee et al., 2015). We found detailed discussions related to Blockchain Technology (e.g., the characteristics, the different challenges, and opportunities related to Blockchain Technology) towards possible uses in Industry 4.0 were presented in Viriyasitavat and Hoonsopon, (2019), Lu (2019), Viriyasitavat et al. (2019), Zheng et al. (2017), Mohamed and Al-Jaroodi (2019).

We found that Blockchain acts as "an immutable ledger" (Zheng et al., 2017) which consents for the associated transactions to be placed in a decentralized approach in Industry 4.0 scenario. A detailed discussion related to the architecture of Blockchain, consensus, and its future was further presented in (Zheng et al., 2017). In

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another research, the integration of Blockchain and service-oriented middleware (e.g., Man4Ware) was presented in (Mohamed and Al-Jaroodi, 2019) to facilitate several smart manufacturing activities e.g., orders for materials, shipment, energy, manufacturing procedures, machine activities, activities related to safety, research & development data, monitoring activities, storage logs, payments, etc. Though several applications of Blockchain technology may be found in literature, we particularly got interested in the Blockchain consensus, which we believe to be very important for the construction of a Blockchain. We found a fresh consensus PoR (Proof of Reputation) was presented as an enhancement of Delegated Proof of Stake (DPoS) to empower the existing limitation in the scalability of PoW (Proof of Work) in the Blockchain network since the existence of Bitcoin (Do. et al., 2019). In another work, the three factors-based i.e., asset, transaction activity, and consensus participation-based proof of reputation consensus mechanism was presented in (Zhuang et al., 2019). It was further presented in Viriyasitavat et al. (2019) that scalability and security at an acceptable decentralization were achieved for the presented system while replacing "coin-staking" (Do. et al., 2019) based on several ranking theories e.g., HodgeRank (Jiang et al., 2011), NCDawareRank (Nikolakopoulos and Garofalakis, 2013), and PageRank (Langville and Meyer, 2004). As ranking theories and their applications are important in Blockchains, a brief discussion on said ranking theories (mostly PageRanks) is presented next.

PageRank plays a significant role in ordering relevant web pages based on the relevancy of user queries. Several researchers have focused on the enhancements of PageRank computation. By convention, low-va PageRank indicates high PageRank (Nikolakopoulos and Garofalakis, 2013; Langville and Meyer, 2004; Kundu et al., 2007; Guha et al., 2015; Guha et al., 2015). Detailed discussions on the effective role of PageRank towards business growth and profits i.e., SEM (Search Engine Marketing) were presented in (Aswani et al., 2018, Aswani et al., 2021). Besides, the failures in SEM and their effects were explored in (Aswani et al., 2018) along with a possible way to decide and avoid such situations by detecting and avoiding spam websites (Aswani et al., 2021). Several other approaches to facilitating the automated decision-making process in several real-life scenarios may also be found in Duan et al. (2019), Akter et al. (2021), Ziolkowski et al. (2020). Several static/dynamic PageRank computations may be found in different scientific databases to facilitate an effective PageRank computation. Among several others, PageRank computations presented in Kundu et al. (2007), Guha et al. (2015), Guha et al. (2015) were particularly of great interest.

Besides PageRank, several other ranking approaches were also investigated and presented by researchers. For example, HodgeRanks was presented in (Jiang et al., 2011), which investigated pairwise ranking denoted as edge flows in graphs involving discrete/combinational Hodge Theory on imbalanced and incomplete datasets obtained from the Internet, along with a "certificate of reliability" (Jiang et al., 2011); NCD (Nearly Completely Decomposable) system was presented (Nikolakopoulos and Garofalakis, 2013) and an NCDawareRank was presented to "generalize and refine PageRank" (Nikolakopoulos and Garofalakis, 2013) in Nikolakopoulos and Garofalakis (2013). Besides, discussions on HodgeRank, and NCDawareRank, we found discussions on trust and or, organizational value creation were presented in Liang et al. (2021), Chen et al. (2022), Schlatt et al. (2022), Du et al. (2019), Gol et al. (2019) concerning enhancements in Blockchains. From our studies, we particularly learned that all such computations e.g., HodgeRank,lued and NCDawareRank have scope toward enhancements for Blockchains.

It has already been discussed the significance of PageRank in view of SEM and or Blockchains. Thus, effective modeling of PageRank and related validation should have its importance. In Kundu et al. (2007), we found uses of the Cellular Automata (CAs) (Wolfram, 2002, Chaudhuri et al., 1997) technique for the dynamic modeling of PageRank. In another series of research, energy-based dynamic modeling of PageRank was presented in Guha et al. (2015), Guha et al. (2016) to enhance the PageRank computation. Though effective PageRank computation is a focus among researchers over time, we found that validation of computed PageRank is an equally important research area and has received significant attention over time. CAs-based PageRank validation (i.e., an extended PageRank) algorithm was first introduced in Mitra and Kundu (2015). Thereafter, concerning the energy-based modeling of PageRank in (Guha et al., 2015, Guha et al., 2016), the energy-based modeling of PageRank validation was presented in Mitra and Kundu (2017). Detailed energy-related characteristics for said validation model were further investigated in Mitra and Kundu (2017), Mitra (2019) and concluded the presence of energy stability in such validations (Mitra, 2019). Further, it was concluded that the Power-law distribution of web data (Mitra, 2021) was preserved in CAs-based modeling of PageRank validation. It may be noted that past researchers focused on the uses of CAs as CAs-based physical models may easily be incorporated with VLSI (Very Large-Scale Integration) design at the cost of D-FFs (Flip-flops) with inherent parallel computing capability (Chaudhuri et al., 1997). For this reason, we decided to continue with CAs-based modeling and validation of PageRank. Hence, a brief description of CAs is presented next.

CAs (Wolfram, 2002, Chaudhuri et al., 1997) are a popular choice among researchers for modeling Engineering/scientific problems, those are inherently dynamic and complex in nature. In CAs-configuration, cells are considered as basic units of CAs arrangement in the form of a lattice. CAs arrangement may be found in single or multi-dimension scenarios. Such configuration evolves over space and time (both are discrete in nature) based on CAs rule(s). The simple CAs configuration is found in one dimension with a 3-neighborhood (dependencies with a left neighbor, self (self-cell), and right neighbor) at PB (Periodic Boundary) or, NB (Null Boundary) scenario, and is popularly known as elementary CAs (ECAs)). Reader(s) of this article is(are) sincerely requested to go through (Chaudhuri et al., 1997) to have more information on ECAs.

The next state of ECAs may be achieved with the following by Equation 1 (Chaudhuri et al., 1997).

$$S_i^{t+1} = f_i\left(S_{i-1}^t, S_i^t, S_{i+1}^t\right) \tag{1}$$

Where, the next state function (rule) is presented by 'f_i'; current state value of 'i^{th}' cell at a time 't' is presented by 'S_i^t'; current state value of '$i-1^{th}$' cell at a time 't' is presented by 'S_{i-1}^t'; current state value of '$i+1^{th}$' cell at a time 'a t' is presented by 'S_{i+1}^t' and state value of 'i^{th}' cell at time '$(t+1)$' is presented by 'S_i^{t+1}'.

On the other hand, environmental sustainability and Green computing are gradually becoming a new pre-requisite conditions to be fulfilled towards the deployment of Engineering works. For this reason, several researchers have focused on the design and development of low-power-consuming models. Among several works facilitating Green Computing, we found research focuses were also intended to facilitate an energy-efficient and low-power-consuming PageRank validation (Mitra and Kundu, 2015, Mitra and Kundu, 2017, Mitra, 2019, Mitra, 2021). A brief discussion on energy-related focus in CAs-based PageRank validation as discussed in (Mitra and Kundu, 2015, Mitra and Kundu, 2017, Mitra, 2019, Mitra, 2021) is further presented in Section 3. We further observed that exploration of Green Computing was never limited to PageRank / PageRank validation. Among several others, we found recent research has focused on Green Blockchain (Imbault et al., 2017). For this reason, we found scope for research in Green Blockchains.

We studied that the "design of the consensus mechanism strongly impacts the blockchain system performance, including transaction capacity, scalability, and security" (Zhuang et al., 2019). In our studies, we found that the DPoR (Delegated Proof of Reputation) consensus design in Blockchain Trilemma (that emphasized a discussion about "...trade-off among "security, decentralization and scalability"." (Do et al., 2019)) relies on PageRank, NCDawareRank, and HodgeRanks (Do. et al., 2019). Additionally, in Do et al. (2019), Do et al. suggested for a replacement of stake voting with reputation voting to enhance Blockchain Trilemma. In Do et al. (2019), it was thematically described that DPoR-based consen-

sus design in Blockchain Technology enjoys energy efficiency, (semi-) decentralization, high scalability, staking, reputation credit, voting, high fairness & balance, and a high incentive for node honesty and responsibility as compared to the PoW, PoS, DPoS, and PoI. Unfortunately, we found validation of PageRank was not considered in Do et al. (2019), which we believe to be an effective simplification and a possible enhancement toward the consensus design in Blockchain Technology. We found that research focused on energy-efficient Blockchain consensus is still relevant in the present era and further has high potential in view of Industry 4.0 uses. For this reason, we choose to investigate the scope for PageRank validation toward DPoR-based consensus design in Blockchain Technology, which may further offer a Green design for possible uses in Industry 4.0 scenario.

The major findings of the presented research are enlisted below.

i An enhanced consensus in Blockchain Technology for Industry 4.0 uses is presented.
ii A new consensus for DPoR-based Blockchain is designed by extending PageRank algorithm.
iii Uses of CAs-based PageRank validation model is emphasized.
iv A new algorithm is presented for a cost-effective computation of reputation score.
v Low energy consumption is ensured in presented model.
vi Detailed dynamics are examined to explore its cost-effective modeling capability.

The architecture of the remaining paper is as followed. Section 2 presents related works, Section 3 briefly describes the background works, Section 4 presents proposed work; Section 5 presents result discussions, and Section 6 presents the concluding remarks and future research direction.

2. Related Works

A survey of the state-of-the-art literature from several scientific databases may explore the popularity and impact of Industry 4.0 and related research areas. Several hundred or probably thousands of research papers might be retrieved from any scientific database. For this reason, it is ridiculous to present all published pieces in a single document. For this reason, we carefully restricted our literature reviews to a limited number of resources, those having a significant role in our presented work.

A brief discussion on the focus, methodology, and future direction of research related to Industry 4.0, has already been presented in Section 1. Besides, we observed that the roles of Blockchain Technology in smart manufacturing lines were introduced in different distinct efforts by researchers (Lee et al., 2015, Mitra, 2021, Ahsan and Siddique, 2022, Zheng et al., 2017, Mohamed and Al-Jaroodi, 2019, Do et al., 2019). Thus, we believe that there exists a scope for further investigation towards the possible enhancements of Industry 4.0 by involving an enhanced Blockchain Technology. For this reason, a brief discussion on the past important efforts towards the enhancements of Blockchains is presented next.

We found a discussion in Do et al. (2019) regarding the roles of PoR, DPoS, and PoW towards enhancements of the Blockchain networks. We further studied that the reputation ranking of Do et al. (2019) successfully replaced pure coin-staking with a system based on HodgeRank, NCDawareRank, and PageRank. We observed that beyond Blockchain, the reputation ranking of Do et al. (2019) is also capable to be used for checking credit, social reputation, and rating. In a different approach, the Proof of Luck (PoL)-based consensus protocol was introduced in Milutinovic et al. (2016) to enhance the Blockchains. In Milutinovic et al. (2016) we found that TEE platform-based random number generation was chosen as a consensus leader to facilitate transaction validation at low latency, low deterministic confirmation time, very low amount (almost negligible) of energy consumption, and "equitably distributed mining" (Milutinovic et al., 2016).

Please note that we had already mentioned that our interest lies in the research towards modeling of Green Blockchains (refer to Section 1). For this reason, we studied several research papers targeting Green Blockchains. Among others, we found literature on Blockchain technology that was applied on Predix (an Industrial operating system) in view of Green Certificates within an eco-district (Imbault et al., 2017). In another distinct attempt, to boost cost-effective participation from customers, an access control mechanism was presented in Samuel et al. (2019) by offsetting customers for their involvement in data sharing via blockchain at differential privacy. It was additionally claimed in Samuel et al. (2019) that the presented approach of (Samuel et al., 2019) was efficient with reference to minimized privacy risk, maximized aggregator profit, and low computational cost. To enhance the Blockchains, another prototype was presented in Xue et al. (2018) by adapting the PageRank algorithm to assess the participation and significance of the users. Thereafter we found that an alternative model was presented in Sharma (2018) to handle the energy demands of the blockchain-enabled scenario by optimally regulating the number of transactions through distributed clustering. It was further found about the performance of the proposed approach (Sharma, 2018) that "40.16% better in terms of energy conservation and 82.06% better in terms of the number of transactions required to share the entire blockchain data compared with the traditional blockchain" (Sharma, 2018). In a different approach, CoT (Circle of Trust) was introduced in (Ravindran, 2019) to overcome the existing flaws of PoW, PoS and DPos. Additionally, CoT-based architecture was presented as "capable of scaling the transactions per second, as the network of miners grows, which no other blockchain on the market is capable of doing" (Ravindran, 2019).

We observe that since its introduction and further enhancements of Blockchains, they became quite popular among technology users due to their enhanced security and privacy features. We observe that several commercial and engineering applications with Blockchains have been presented by researchers as a solution for enhanced data security and privacy. For this reason, a brief description of recent Blockchain Technology-related literature in view of today's business and or Industry 4.0 uses, is presented next.

In our studies, we found a literature in Grover et al. (2018) that examined the true potential for Blockchains in view of different business functions e.g., facilitating instant payments, building trusted interfaces, and maintaining product traceability in different business scenarios (i.e., B2(business to)B(business), B2C(customers), and B2G(governments)). We found, the truly multidisciplinary nature of blockchain technology, as examined in (Ghosh, 2019), may successfully be used in several real-life applications. In Prakash et al. (2022), Amponsah et al. (2022), Jain et al. (2021), Mittal et al. (2021), we observed several opportunities for Blockchains were investigated for their possible consideration in cutting-edge Information Systems e.g., for possible improvement of financial security in health insurance (Prakash et al., 2022), enhanced marketing policy (Amponsah et al., 2022), for possible enhancements in cyber security (Jain et al., 2021) and healthcare (Mittal et al., 2021) scenarios, for possible inclusion in banking scenarios (Chen et al., 2022), etc. Additionally, we observe that the application suitability for Blockchains in view of Industry 4.0 scenarios was investigated in two recent efforts (Javaid et al., 2021, Zuo, 2021) along with a possible future research direction toward enhancements for Blockchains (Zheng and Lu, 2021) over its drawbacks (Abu-Elezz et al., 2020).

From our studies (Javaid, Haleem, Singh, Khan, & Suman, 2021; Prakash, Anoop, & Asharaf, 2022, Amponsah et al., 2022, Jain et al., 2021), we learned that Blockchain technology facilitates stable and effective data sharing, with the advantage of the incorporation of an "unchangeable database of all messages exchanged by various connected smart devices is established" (Javaid et al., 2021). Additionally, identity protection may also be achieved with the incorporation of this technology. Due to its huge potential for possible uses in trustworthy and secure applications, we observe that several researchers feel an urgent

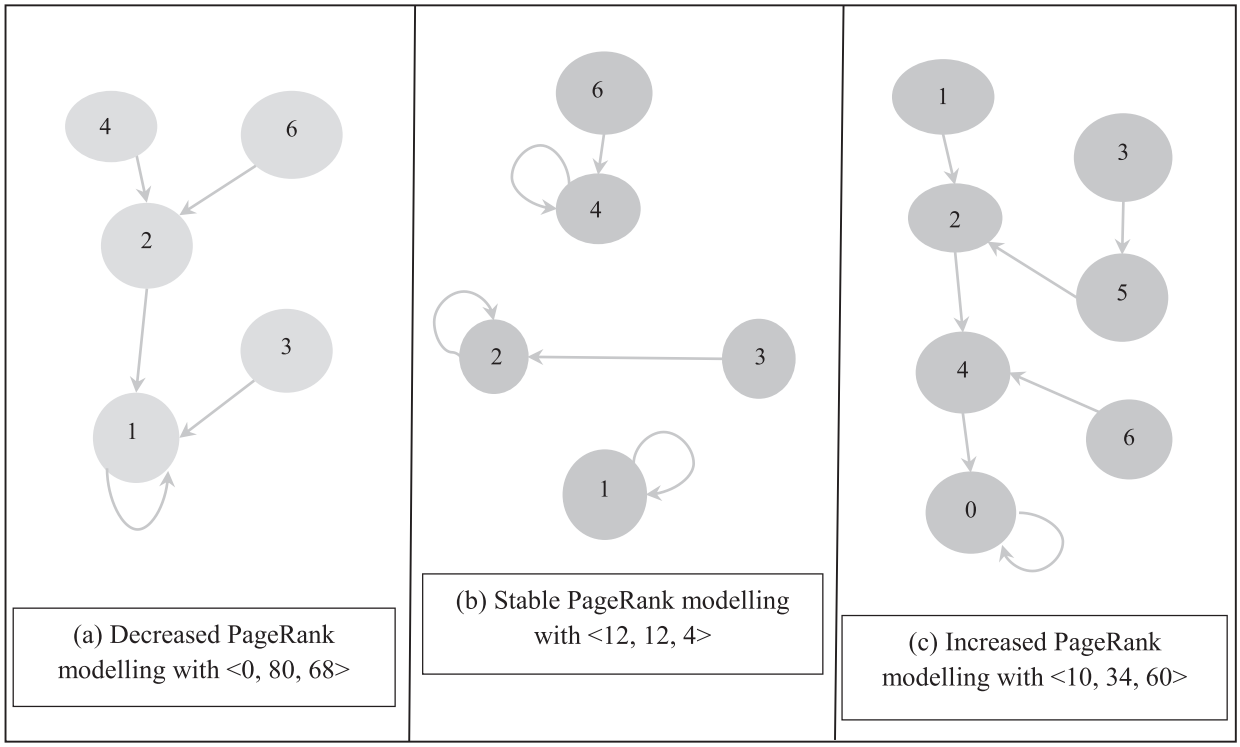

Fig. 1. Transition diagram for ECAs-based PageRank validation model (influenced from Mitra and Kundu, 2015).

need to enhance the existing technology to make it a more cost-effective and secure technology. Thus, we feel the urge to continue our effort to enhance the existing technology, if possible.

Earlier, we already found that PageRank and PageRank validation play a significant role in the design of Blockchain consensus (Do. et al., 2019, Milutinovic et al., 2016, Samuel et al., 2019, Xue et al., 2018, Sharma, 2018, Ravindran, 2019). We also found that PageRank validation-based approaches are better suitable to detect possible intentional manipulation in PageRank and or forgery in computed PageRank[34-37]. Thus, we chose to focus on PageRank validation-based possible enhancement(s) in Blockchain Technology. As our presented research is greatly influenced by the existing PageRank validation technology, we present a brief discussion on the available literature(s) on PageRank validation in Section 3.

3. Background

An ECAs-based model in the NB scenario with only three CAs cells was presented in (Mitra and Kundu, 2015) for the validation of PageRank against possible manipulation of PageRank changes and spamming. A brief discussion on said model is presented next.

Different scenarios in the case of dynamic PageRank validation were modularized into i) stable traffic (was modeled with CAs rule combination 12, 12, 4), ii) decreased traffic (was modeled with CA rule combination 0, 80, 68), and iii) increased traffic (was modeled with CAs rule combination 10, 34, 60). The CAs-based transition diagram of (Mitra and Kundu, 2015) is presented once again in Fig. 1.

Let us follow the notations to recognize Fig. 1. Circular spaces imply states, arrow implies transitions between state(s), and number implies state number. $\langle , \rangle$ implies a set of CAs rules (wolfram codes) accountable for the creation of concerned transition diagrams.

For an n-cell ECA scenario, there exists a total of 2^n number of states, i.e., for a 3-cell ECAs formation, there exists a total of 8 states (state 0 to state 7). State 0, and state 7 were not considered in Fig. 1 as they were unacceptable states with respect to PageRank validation (Mitra and Kundu, 2015). The reader(s) of this paper is(are) requested to go through (Mitra and Kundu, 2015) for a detailed insight into the design philosophy for the CAs-based PageRank validation model.

In another research, energy requirements by CAs-based models were studied (Mitra and Kundu, 2017). It was presented in (Mitra and Kundu, 2017) that power expenditure for D-FFs (i.e., CAs cells) was very low; maximum to minimum power consumption for FFs was $1.20E-05$ watts to $1.17E-07$ watts at different CMOS technologies. A full discussion associated with the energy requirement by CAs (D-FFs) may be further found in (Mitra and Kundu, 2017).

In a different research, ECAs dynamics at different boundary scenarios were examined in Grover et al. (2018), Ghosh (2019), Prakash et al. (2022) to explore its true potential for cost-effective modeling. Unfortunately, in Mitra and Kundu (2015), Mitra and Kundu (2017), Mitra (2019), Mitra (2021), we have not found such investigations towards CAs-based PageRank validation. For this reason, we continued our focus on the CAs-based PageRank validation model at different fixed boundary scenarios for its possible uses in Blockchain consensus. Initially, the CAs-based PageRank validation model in the NB scenario of (Mitra and Kundu, 2015) is considered in our presented work. Further, we presented a detailed investigation of the dynamics of such a model in several fixed boundary scenarios.

4. Proposed work

We found that the reputation concept in Do et al. (2019) was a function of three different factors: i) staked amount, ii) resource usage, and iii) transaction activity. The reputation score as determined in Do et al. (2019) is once again presented in Equation 2.

$$Reputation\ Score = f\ (P,\ U,\ R) \tag{2}$$

where, P denotes stake power (based on stake coins); U denotes resource (CPU, RAM, bandwidth) usages; R denotes transactions-based-reputation rating involving active accounts with many in/out transactions (similar to web links).

Equation 3 together with P (normalized stake power), U (resource usage), and R (rating score) finally computed the $Reputation\ Score$ using

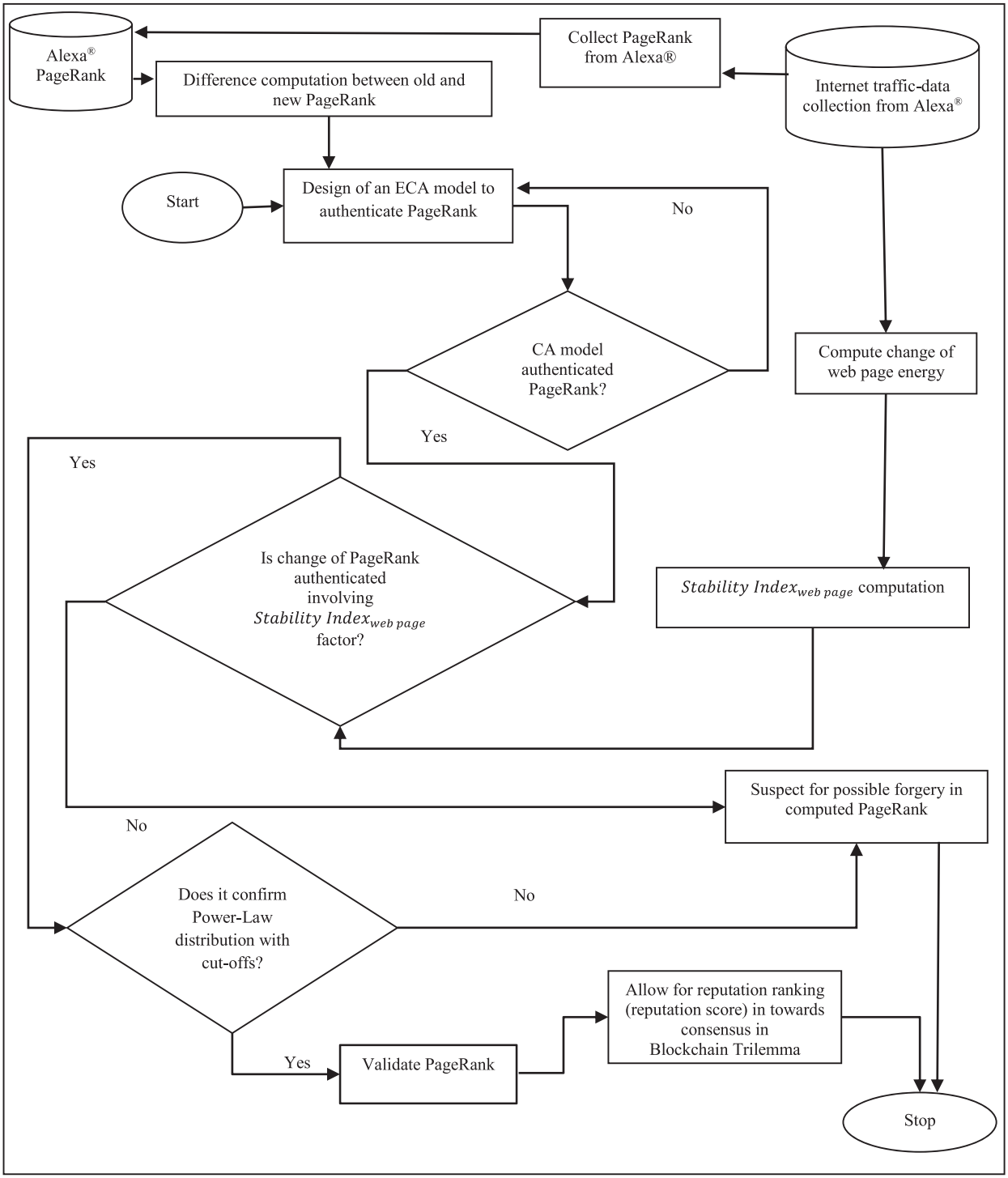

Fig. 2. Projected system towards consensus in Blockchain Trilemma involving $Stability\ Index_{web\ page}$ aware PageRank validation model in CA (influenced from Mitra and Kundu, 2017, Mitra, 2019, Mitra, 2021).

Equation 3 (Do. et al., 2019).

$$Reputation\ Score = \left(c_1 P + c_2 U + c_3 R\right) \tag{3}$$

where, for weight $c_i\ \ for\ \ i = 1, 2\ \ and\ \ 3,\ \ c_1 + c_2 + c_3 = 1$ (refer to (Do. et al., 2019)).

Though there exist several distinguishing features between Web and Blockchain networks but the links in both networks are kind of similar types of connections (Do. et al., 2019). Thus, considering the similar features between Web and Blockchain networks, we propose the inclusion of the energy-stability-based PageRank validation model towards the consensus in DPoR-based Blockchain Trilemma as energy-stability-based PageRank validation is an advanced and dynamic model where simple connection-based hyperlinks are not considered towards the validation. Thus, we find our proposed system overcomes the distinguishing features between simple Web and Blockchain networks as presented in (Viriyasitavat et al., 2019) and we believe that the proposed approach has a complete potential for use in Blockchains.

We considered the energy-based PageRank validation model of (Ziolkowski et al., 2020). The computation of change of page energy

Table 1
Detailed Internet traffic data collected with Alexa®.

URL	Bounce Rate (%)		Daily page views per user		Total time Spent (in second)		Total number of In-links		Average load time (in second)		Global PageRank achieved from Alexa®		Remark
	Date 27.01.2018	*Date 01.07.2020*	*Date 27.01.2018*	*Date 01.07.2020*	*Date 27.01.2018*	*Date 01.07.2020*	*Date 27.01.2018*	*Date 01.07.2020*	*Date 27.01.2018*	*Date 01.07.2020*	*Date 27.01.2018*	*Date 01.07.2020*	
"flipkart.com"	43.4	27.2	3.73	6.55	356	470	16204	7518	2.878	1.953	160	144	Increased PageRank and can be validate using Figure 1(c)
"news18.com"	66.3	53.3	1.74	2.79	189	219	4343	1612	2.518	2.238	1509	611	

may be performed using Equation 4 (Ziolkowski et al., 2020).

$$\Delta\ E = E_{web\ page_{new}}\ \sim\ E_{web\ page_{old}} \tag{4}$$

where,
$E_{web\ page} = (|bounce_rate| + |daily_page_views| + |time_spent| + |in_links|)$

The "$Stability\ Index_{web\ page}$" of (Mitra, 2019) towards the validation of PageRank changes in view of energy stability is once again considered in the present research. Computation of "$Stability\ Index_{web\ page}$" towards PageRank validation may be obtained using following Equation 5, which involved Equation 4.

$$Stability\ Index_{web\ page} = |(percentage\ of\ change\ in\ page\ energy \times change\ in\ average\ load\ time)| \tag{5}$$

$$Stability\ Index_{web\ page} = \begin{cases} 0\ (or,\ almost\ zero) \\ small\ valued \\ large\ valued \end{cases} \tag{6}$$

We observed that the three different values in Equation 3 represents three different PageRank scenarios: stable (no changed) PageRank (refer to 0(or, almost zero) of Equation 6), increased PageRank (refer to small valued of Equation 6), and decreased PageRank (refer to large valued of Equation 6) respectively. Thus, we chose to use Equation 6 in the proposed approach. Further, to ensure additional trustworthiness, an additional level of checking involving the satisfiability of Power-Law distribution with cut-offs as presented in (Mitra, 2021) is also considered in our proposal.

The flowchart of the projected system towards DPoR-based consensus in Blockchain Trilemma involving $Stability\ Index_{web\ page}$ aware PageRank validation model in CAs, is shown in |

Figure 2. It may be noted that the presented design simplifies the computation of *Reputation Score* as it uses an advanced approach towards validation of PageRank instead of considering three different ranks (e.g., PageRank, HodgeRank and NCDawareRank) (refer to (Do. et al., 2019)) and offers an enhanced level of reliability against possible forgery/manipulation in PageRank including spamming.

Following Algorithm 1 was designed to facilitate projected system flow in Figure 2.

Algorithm 1
Computation of reputation score involving energy-stability-based validation of PageRank.

Input: web-traffic data.
Output: reputation score.
Step 1. Start
Step 2. Compute stability_index using Equation 4
Step 3. If stability_index ≈0, follow Step 6 else Step 12
Step 4. Else if stability_index ≈high valued number, follow Step 7 else Step 12
Step 5. Else if stability_index ≈low valued number, follow Step 8 else Step 12
Step 6. If is it reinforced by CA based design of Figure 1 (b), follow Step 9 else Step 12
Step 7. If is it reinforced by CA based design of Figure 1 (c), follow Step 9 else Step 12
Step 8. If is it reinforced by CA based design of Figure 1 (a), follow Step 9 else Step 12
Step 9. If is it reinforced by Power_Law distribution, follow Step 10, else Step 12
Step 10. Authenticate change of PageRank
Step 11. Compute reputation rank using Equation 3, follow Step 13
Step 12. Nullify change of PageRank and suspect for possible forgery, follow Step 13
Step 13. Stop.

Computation of reputation score based on our approach using Equation 3 is presented next section.

Table 2
Computation of Stability Index based on Table 1 data

URL	Page energy		Average load time (in second)		Change in Stability Index parameter as per Equation 6	Remarks based on Power-Law distribution	Conclusion
	Date 27.01.2018	*Date 01.07.2020*	*Date 27.01.2018*	*Date 01.07.2020*			
"flipkart.com"	16607.13	8021.75	2.878	1.953	0.4782	Ensured Power-Law distribution with cut-offs	No manipulation in change of PageRank is observed
"news18.com"	4600.04	1887.09	2.518	2.238	0.1651		

Table 3
Computation of reputation score using Equation 7.

URL	Reputation score	
	At initial perspective	*At long-term perspective*
"flipkart.com"	0.6 θ+0.1913	0.6 θ+0.2869
"news18.com"	0.6 θ+0.066	0.6 θ+0.099

5. Results and discussions

To analyze the performance for the proposed approach we collected several real time web-data using free Internet traffic measurement tool. Collected data and related discussions are systemically presented in following subsections.

5.1. Data collections

We collected Alexa®-based real-time Internet traffic data. A detailed discussion on energy related studies towards several URLs was already presented in (Mitra and Kundu, 2017, Mitra, 2019, Mitra, 2021). For simplicity, we reported web traffics for few URLs in Table 1 to investigate further.

Please recall that, low valued PageRank indicates high PageRank. For this reason, we observed increased PageRank over time for both URLs. Computation of $Stability\ Index_{web\ page}$ for said URLs is presented in following Subsection 5.2.

5.2. Data analysis

Alexa® collected real time traffic data (refer to Table 1) is further analyzed to find out its stability (if any). Computation of $Stability\ Index_{web\ page}$ for said URLs (refer to Table 1) is presented in following Table 2.

Computation of reputation score as presented in Do et al. (2019) described that a "blockchain network does have few transactions in the beginning stage, developers can set a dominant rate for stake power, for example, $c_1 = 0.8$, $c_2 = 0.2$, $c_3 = 0$. Months later may be $c_1 = 0.6$, $c_2 = 0.3$, $c_3 = 0.1$. A suggestion for long-term perspective is $c_1 = 0.4$, $c_2 = c_3 = 0.3$" (Do. et al., 2019), and threshold (θ) for P (stake power) where, $\theta \approx \$100 \pm 10$. On the other hand, we found in (Mitra, 2019) that validation for change of PageRank considers resource uses and further it is directly related to change of $Stability\ Index_{web\ page}$. For this reason, we used $Stability\ Index_{web\ page}$ towards R (rating score) and U (resource usage) in Equation 3. Thus, we find scope for c_2 and c_3 do not exists further and it simplifies Equation 3. Computed reputation scores are presented in following Table 3 using proposed Equation 7, which is a simplified version of Equation 3; we followed the same parameter values for computation of reputation scores for several URLs (refer to Table 2).

$$Reputation\ Score = \left(c_1 P + \left(1 - c_1\right)\ Stability\ Index_{web\ page}\right) \tag{7}$$

By considering 0.6 θ of Table 3 as a constant value, we further presented the characteristics of reputation scores at initial and long-term perspective in following Fig. 3.

It is observed from Fig. 3 that a linear growth trendline is achieved for reputation score for each of the URLs with reference to the reputation scores at initial and long-term perspective. Reputation score for both URLs increases over time. Please recall that both URLs showed increased PageRank (refer to Table 1). Hence, we believe that achieved result is highly favorable towards its possible consideration for uses in Blockchain network as it ensures a linear growth over time. It may be noted that as our present focus is on the possible enhancement(s) for reputation score in view of Blockchain consensus, we restricted our present investigations. Investigations related to Blockchain performance evaluation with reference to several parameters e.g., transactions per second (TPS), delay, etc. have not been considered in present study.

Towards the discussion on robustness for the proposed approach, a comparative study with reference to three existing approaches (e.g., PageRank, MCDawareRank, HodgeRank) is presented in following Section 5.3.

5.3. Robustness checks

It may be concluded from Section 5.2 that presented PageRank validation model is based on the changes in PageRank (if any). Thus, we find that Its characteristics are similar to those of PageRank. A detailed comparison among PageRank, MCDawareRank, HodgeRank were presented in Do et al. (2019). We extended our study based on Do et al. (2019) and presented our findings in following Table 4.

In Table 4, 'N' denotes 'No' and 'Y' denotes 'Yes'. It is observed from Table 4 that validation of PageRank validation model is achieved at lowest complexity rate as compared to NCDawareRank, and HodgeRank while ensuring both the validation and spamming resistance. Conclusive findings based on the comparison and its further analysis is presented in following Section 5.4.

5.4. Contributions to literature

From our studies, we observed that PageRank validation-based approaches were found to be better suitable to detect possible intentional manipulation of PageRank in dynamic scenarios (refer to Section 2) as compared to PageRank-based approach (i.e., without validation) (Mitra and Kundu, 2015, Mitra and Kundu, 2017, Mitra, 2019, Mitra, 2021). Additionally, from our present investigations, it is observed from Table 4 (refer to Section 5.3) that PageRank validation model might be used in ranking theory as it is found to be better as compared to PageRank, NCDawareRank, and HodgeRank. It may further be noted that the proposed PageRank validation-based design is capable of being physically modelled with ECAs, that ensures a low-cost physical implementation, and a requirement for very low energy consumption (i.e., in the range of $1.20E-05$ watts to $1.17E-07$ watts only) (Mitra and Kundu, 2015, Mitra and Kundu, 2017, Mitra, 2019, Mitra, 2021). For said reasons, it may be concluded that PageRank validation-based model is an effective choice towards reputation score.

5.5. Implications for practice

In our studies, we further found that CAs dynamics may change at several fixed boundary conditions (Mitra, 2021, Teodorescu, 2015).

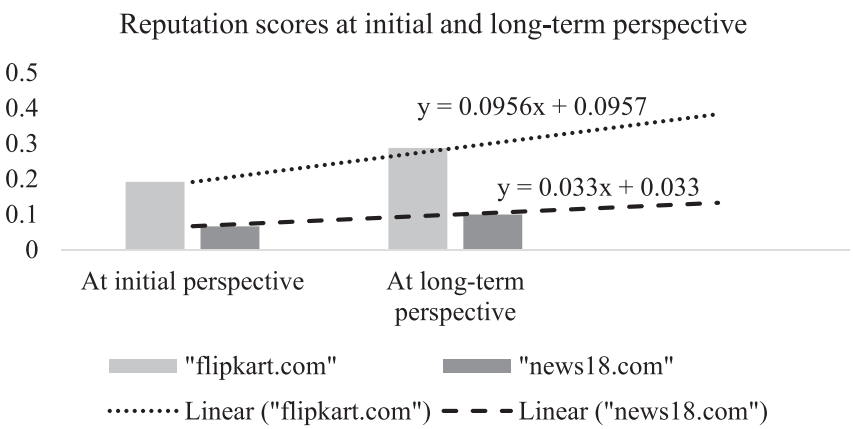

Fig. 3. Characteristics for reputation scores at initial and long-term perspective.

Table 4
Comparison among different ranking theories.

	PageRank (Do. et al., 2019)	PageRank validation	NCDawareRank (Do. et al., 2019)	HodgeRank (Do. et al., 2019)
Proximity matrix	N	N	Y	N
Complexity rate	X	X	Y < X	Z < Y
Validation	N	Y	N	Y
Spamming resistance	N	Y	Y	Y

Table 5
ECAs dynamics at several fixed boundary scenarios for CAs-based PageRank validation.

	0...0	0...1	1...0	1...1
<0,80,68> for decreased PageRank	0→0	0→0	0→0	0→0
	1→1	1→0	1→1	1→0
	2→0	2→0	2→0	2→0
	3→1	3→0	3→1	3→0
	4→2	4→2	4→2	4→2
	5→1	5→0	5→1	5→0
	6→2	6→2	6→2	6→2
	7→1	7→0	7→1	7→0
<12, 12, 4> for stable PageRank	0→0	0→0	0→0	0→0
	1→1	1→0	1→1	1→0
	2→2	2→2	2→2	2→2
	3→2	3→2	3→2	3→2
	4→4	4→4	4→0	4→0
	5→5	5→4	5→1	5→0
	6→4	6→4	6→0	6→0
	7→4	7→4	7→0	7→0
<10, 34, 60> for increased PageRank	0→0	0→0	0→0	0→0
	1→3→4→0	1→3→4→0	1→3→0	1→3→0
	2→5→3	2→5→3	2→1	2→1
	6→5	6→5	4→0	4→0
	7→4	7→4	5→3	5→3
			6→1	6→1
			7→0	7→0

For this reason, we investigated the ECAs dynamics further at different fixed boundary configurations. We followed the identical notations towards the representation for different fixed boundary conditions: 0...0 (i.e., NB), 0...1, 1...0, and 1...1 fixed boundary value respectively at left and right boundary of the ECAs configuration (Mitra, 2021, Teodorescu, 2015). ECAs dynamics as observed in simulations are presented in Table 5.

Let us follow the convention to understand Table 5. Decimal numbers indicate states with state numbers and arrows indicates transitions from source to destination state.

It is observed from Table 5 that ECAs dynamics for PageRank validation model changes at different fixed boundary conditions. Dynamics for CAs-based PageRank validation towards decreased PageRank was found to be identical at 0...0 and 1...0 fixed boundary configurations only, and dynamics for CAs-based PageRank validation towards increased PageRank was found to be identical at 0..0 and 0..1 fixed boundary configurations only. No such identical dynamics was observed for stable PageRank. For this reason, it may be concluded that boundary values for fixed-boundary conditions adversely affect the CAs dynamics towards PageRank validation and hence, CAs-based PageRank validation is required to be operated at 0...0 fixed boundary i.e., NB scenario only. Hence, we may further conclude that though it (physical modelling for CAs-based PageRank validation) requires necessary attention towards fixation of boundary values should results for increased program memory, yet it is capable for being physically implemented with only 3 D-FFs. It also ensures low power consumption (Mitra and Kundu, 2017). For this reason, it may be concluded that it initiates a truly cost-effective design.

6. Conclusions and future research direction

A novel approach towards computation of reputation score using CAs-based PageRank validation at null-boundary scenario is presented in this research. Further, a Power-Law distribution-based investigation in said validation ensured an additional level of checking against any possible manipulation towards the change in PageRank (if any). For this reason, a new algorithm is designed to facilitate $Stability\ Index_{web\ page}$-based reputation score computing which in turn is found to be capable for simplifying an existing computation of reputation score (refer to (Do. et al., 2019)). A comparative study among different ranking theories (refer to Table 4) ensures that validation of PageRank validation model may be achieved at low complexity rate as compared to NCDawareRank, and HodgeRank while ensuring both the validation and spamming resistance. Additionally, proposed CAs-based approach is energy efficient model as CAs-based models ensures energy efficiency (Mitra and Kundu, 2017, Mitra, 2019). Thus, proposed approach is found to be an efficient approach towards the computation of reputation score at Blockchain consensus towards possible uses in Industry 4.0.

To conclude this paper, we would like to highlight that the presented discussion is restricted to provide the theoretic outlines for the computation of reputation score using stability index. A detailed experimental implementations with supporting trials will be presented in our next study with concrete use cases.

Please note, it has already been mentioned that presented study is focused on the possible enhancement(s) for reputation score in view of Blockchains. In future we plan to extend our work and plan to evaluate Blockchain performance with reference to several parameters e.g., TPS, delay, etc.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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